RBG3-MOTION electrical review pack

RackBotGen3, Delta Cup 2026. Spec v0.1.22, repo bac8d89-dirty, built 2026-10-22.

Start with the project overview, then the review brief: it explains the board, what has been verified and the questions. The PDF bookmarks jump to each part.

Part	Source	Page
Project overview	`docs/review/project-overview.md`	2
Review brief and questions	`docs/review/electrical-review.md`	4
Schematic	`pcb/motion-board/fab/rbg3-motion-schematic.pdf`	7
PCB layout (one layer per page)	`pcb/motion-board/fab/rbg3-motion-layout.pdf`	8
3D renders	`pcb/motion-board/fab/*.png`	16
Electrical specification	`docs/spec/02-electrical.md`	18
Motion board README	`pcb/motion-board/README.md`	24
SPICE simulations	`pcb/motion-board/sim/README.md`	27
SPICE results table	`pcb/motion-board/sim/results.md`	32
Datasheet cross-check	`docs/datasheets/CROSSCHECK.md`	33

Source: docs/review/project-overview.md

RackBotGen3: project overview

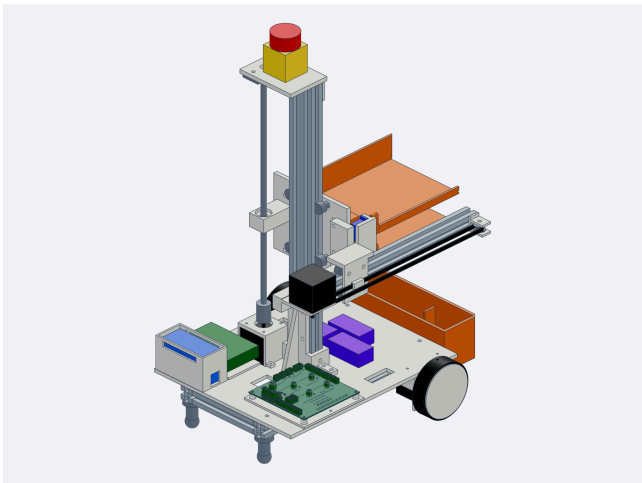
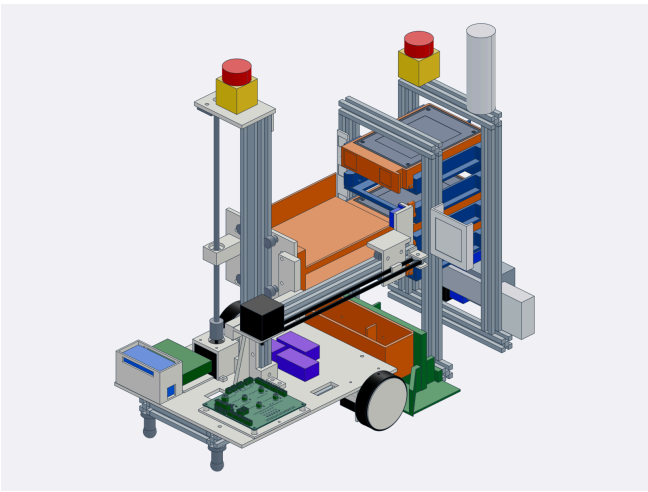
Delta Cup 2026, Automation track. Regional final in Thailand, 14 Nov 2026. Student team of 4.

Objective

A proof-of-concept **autonomous hot-swap robot for data-center drive maintenance**. When the monitoring dashboard reports a failing drive, a small mobile robot, which waits and charges at a spare station:

1. picks a spare 3.5" drive caddy from the spare station;
2. drives to the rack and docks;
3. pulls the failed caddy and pushes the spare into the same bay;
4. drives back to the spare station and puts the failed drive in its RMA slot;
5. stays docked there and charges until the next job.

The pitch is lights-out data centers: no technician trip per failed drive, and a shorter window with a degraded RAID. Delta hardware is inside: a Delta PLC does the rack I/O and safety, and a Delta PSU powers the rack and the charging dock. Target: **≤ 3 min per full swap** and **≥ 95 % success over 50 cycles**.

Robot	Docked at the rack
	

The robot

- **Size.** 300 × 360 mm footprint, ≈ 545 mm tall, ≈ 4.5 kg. Built from 2020/2040 V-slot and 3D-printed parts.
- **Motion.** 4 NEMA17 steppers: 2 direct-drive wheels (differential drive), a Z lift on a lead screw and a Y extractor on a belt. 2 micro servos swing fingers into the caddy handle to pull or push it.
- **Tray.** 2 slots stacked at the bay pitch (45 mm), so one visit to the rack both pulls the failed caddy and inserts the spare.
- **Docking.** A camera finds AprilTags on the rack and spare station; a mechanical V-funnel gives ±2 mm repeatability; spring contacts on the spare-station dock (the robot's home) charge the battery.
- **Compute.** A Raspberry Pi 3 runs vision, the mission logic and the web dashboard. It talks over USB serial to the **motion board**, which generates the steps in real time.
- **Power.** 4S Li-ion pack (≈ 50 Wh, airline carry-on safe) with a BMS, fuse, current monitor and an E-stop relay that cuts motor power.

Where the motion board fits

The motion board (RBG3-MOTION, the subject of this review) is a 2-layer 100 × 100 mm carrier board on the robot base. It holds:

- an ESP32-S3 DevKit (the motion controller);
- 4 TMC2209 stepper driver modules, one per motor;
- connectors for the motors, servos, limit and presence switches, LED, buzzer and current monitor;
- battery-voltage and dock-voltage sensing, and surge protection on the motor supply.

It has no regulators of its own: 5 V and the servo supply come from off-board buck modules. The following pages give the review brief and questions, then the schematic, layout, 3D renders and supporting documents.

Constraints that shaped the design

- About 5 weeks from design to a tested robot (10 Nov 2026), so off-the-shelf modules wherever possible.
- Total budget ≤ 500 USD (estimate ≈ 480 USD).
- Everything must travel as carry-on luggage (7 kg per bag), which also limits the battery to < 100 Wh.
- One-off build: the PCB is ordered from JLCPCB, small surface-mount parts assembled by JLCPCB, and the through-hole parts hand-soldered.

Source: docs/review/electrical-review.md

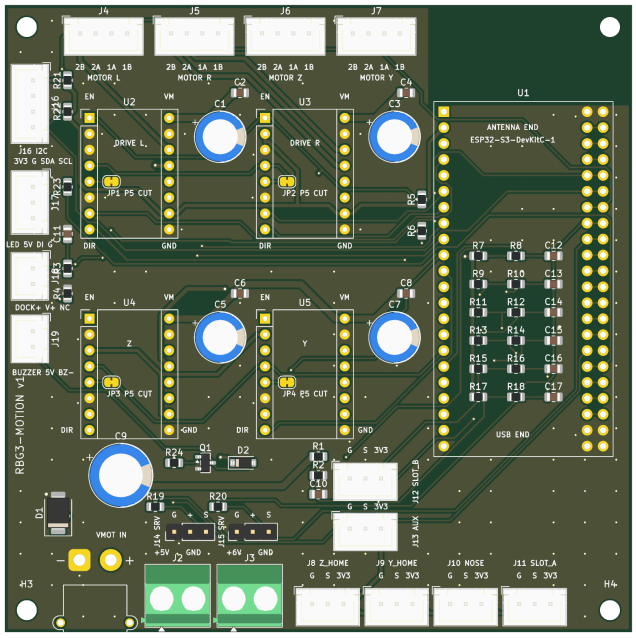
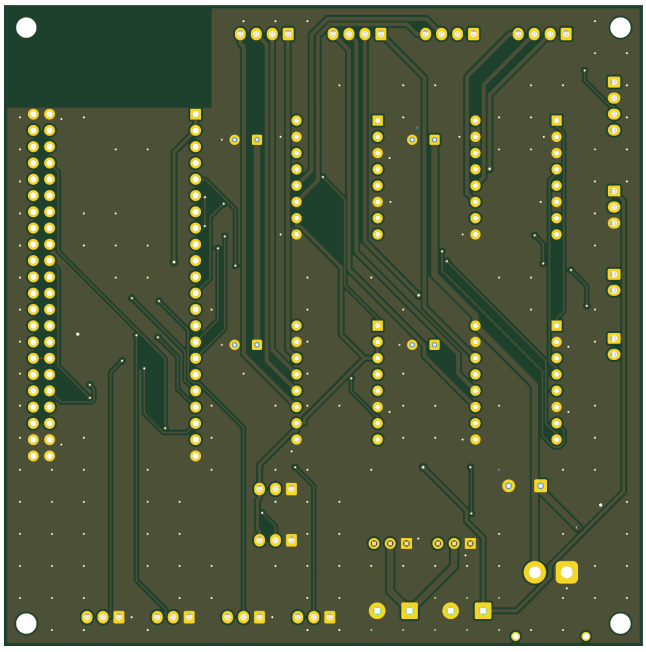
Electrical design review: RackBotGen3 robot (spec v0.1.19)

Thanks for taking a look. This is a student robot for Delta Cup 2026: a small 4-stepper mobile robot that swaps 3.5" drive caddies in a rack. **About an hour of your time is plenty.** The questions below are where a second opinion helps most. Anything else you spot is welcome too.

Timing: the motion PCB goes to JLCPCB (2-layer, 100 × 100 mm, 1 oz) as soon as the review is back, so PCB comments are the most urgent.

What to look at

What	File
Schematic (PDF)	`pcb/motion-board/fab/rbg3-motion-schematic.pdf`
Gerbers (zip) / KiCad 10 project	`fab/rbg3-motion-gerbers.zip` / `pcb/motion-board/`
Board README (connectors, assembly, bring-up)	`pcb/motion-board/README.md`
Electrical spec (power, pin map, connectors)	`docs/spec/02-electrical.md`
Datasheet cross-check already done	`docs/datasheets/CROSSCHECK.md`

Top	Bottom
	

The system in one paragraph

- **Battery.** 4S1P 18650 Li-ion (12.0–16.8 V, ≈ 50 Wh), with a 4S BMS (common port). A 7.5 A blade fuse sits first, at P+, then the main switch, then an INA226 (10 mΩ shunt) feeding the main bus.
- **Charging.** On the dock, 24 V comes in through pads → SS54 → an XL4015 CC/CV module (16.8 V / 1.5 A) → the main bus. The pack charges through the BMS. The dock's – connects only to the charger's IN–, because the module senses current with a shunt between IN– and OUT–.
- **E-stop.** The NC button switches the coil of a 12 V automotive relay, whose contact cuts **VMOT** (steppers and servo buck). Logic stays powered.

- **Motion board.** Purely a carrier: an ESP32-S3-DevKitC-1, 4 × TMC2209 StepStick sockets (one shared single-wire UART, addresses set by MS1/MS2), JST-XH motor and switch connectors, an SMBJ18A TVS and 470 µF bulk capacitance at the VMOT input (XT30), and 100 µF + 100 nF per driver. It has no regulators: 5 V (Pi + LEDs) and 5.8 V (servos) come from off-board buck modules on screw terminals.
- **Power draw.** Idle ≈ 4 W; moving ≈ 12–20 W. Run currents 1.0 / 1.0 / 0.9 / 0.6 A rms per motor.
- **Copper.** VMOT/GND/+6V_SERVO 1.0 mm, motor phases 0.6 mm, 3V3/5V 0.5 mm, signals 0.25 mm, GND pour on both layers.

How this design has been verified so far

1. **KiCad checks** (`pcb/motion-board/scripts/build.sh`, which rebuilds the layout from scratch):

    - ERC: 0 violations.
    - DRC with `--schematic-parity`: 0 unconnected nets and 0 schematic/board mismatches. 19 cosmetic warnings remain (silkscreen overlaps, library footprint mismatches).
    - The autorouter result is kept only with 0 unrouted nets and 0 violations.
    - This shows that the board matches the schematic and meets JLCPCB's clearance rules. It doesn't show that the schematic is right.

2. **Datasheet cross-check** (`CROSSCHECK.md`). Every electrical value in the spec, PCB and firmware was checked against a datasheet stored in the repo, with page numbers. It found and fixed:

    - UART on StepStick pin 5, not 4, on the MKS module;
    - the PSRAM pins are 33–37;
    - fuse 10 A → 7.5 A;
    - ADC full scale 19 V, not 20 V.

3. **SPICE simulations** (`pcb/motion-board/sim/`, ngspice, re-run with `sim/run.sh`). Covered:

    - battery hot-plug onto VMOT;
    - E-stop with the drivers carrying current;
    - charger backfeed with the dock unpowered;
    - the ADC dividers and the UART.

    All pass. Two results need checking on the bench: the inrush (120–220 A for ≈ 100 µs, which might trip the BMS), and the slow E-stop detection when the drivers are disabled.

4. **Hand calculations** in spec 02: power budget, dividers, battery Wh, relay coil.

5. **Reviews by AI agents, not people.** The v0.1.6 "independent review" and a re-check of CROSSCHECK were each done by a separate AI agent with no shared context. **You are the first human reviewer.**

Not done: trace temperature rise (IPC-2152; widths are rules of thumb), any test on the bought modules, and any hardware. Nothing has been built or powered yet. The step-by-step bring-up plan, which stops at the first failure, is in the board README ("Assembly and bring-up checklist").

Questions

Power and safety (spec 02 §1, §5)

1. **Charger topology.** Is it sound to run a CC/CV buck into the main bus, so it feeds the loads and charges the pack through a common-port BMS, while the robot is running? Is anything missing, such as a diode or ideal diode, to stop the bus backfeeding the charger when the dock power is off?

2. **Charger grounding.** Is connecting dock − only to the charger's IN− enough? Is there a failure mode we missed, such as a chassis contact touching the dock frame?
3. **Fuse and BMS.** Is a 7.5 A blade fuse at P+ with a ≥ 15 A BMS reasonable, given that normal draw is under 2 A and the cells are rated 8–10 A max?
4. **E-stop.** Is a relay contact cutting VMOT with the TMC2209s enabled acceptable? Should the firmware disable the drivers first, or is the TVS plus bulk capacitance enough to absorb the motor energy?

Motion PCB (spec 02 §3)

5. **TVS margin.** The SMBJ18A clamps at 29.2 V at full pulse current, which equals the TMC2209's 29 V operating limit (32 V absolute max). Keep it, or move to the SMBJ17A (27.6 V)?
6. **Reverse polarity.** The only protection is the keyed XT30 plus the TVS and fuse; there is no series P-FET. Acceptable for a one-off v1?
7. **Copper widths.** VMOT 1.0 mm at about 4 A peak, and +6V_SERVO 1.0 mm carrying up to 3 A of servo stall current on 1 oz copper. Enough, or widen or pour?
8. **Grounding.** Motor return currents and logic share the GND pours. Does the layout (drivers on the left, ESP32 on the right, VMOT entering bottom-left) look OK?
9. **Decoupling.** Is 100 µF / 35 V + 100 nF per driver plus 470 µF bulk enough at these currents?
10. **UART.** Single-wire UART to 4 drivers: ESP TX through 1 k, RX direct, PDN_UART on pins 4 and 5 (JP1–JP4 bridged). Any concern?
11. **ESP32 pins.** Strapping pins 0/3/45/46 and the octal-PSRAM pins 33–37 are avoided (pin map in spec 02 §3.1). The ADC dividers are 100 k / 18 k for VMOT (≈ 19 V full scale) and the dock sense. Any issue?
12. **Manufacturability.** Footprints, silkscreen or JLCPCB rules: anything that would bite on a first order?

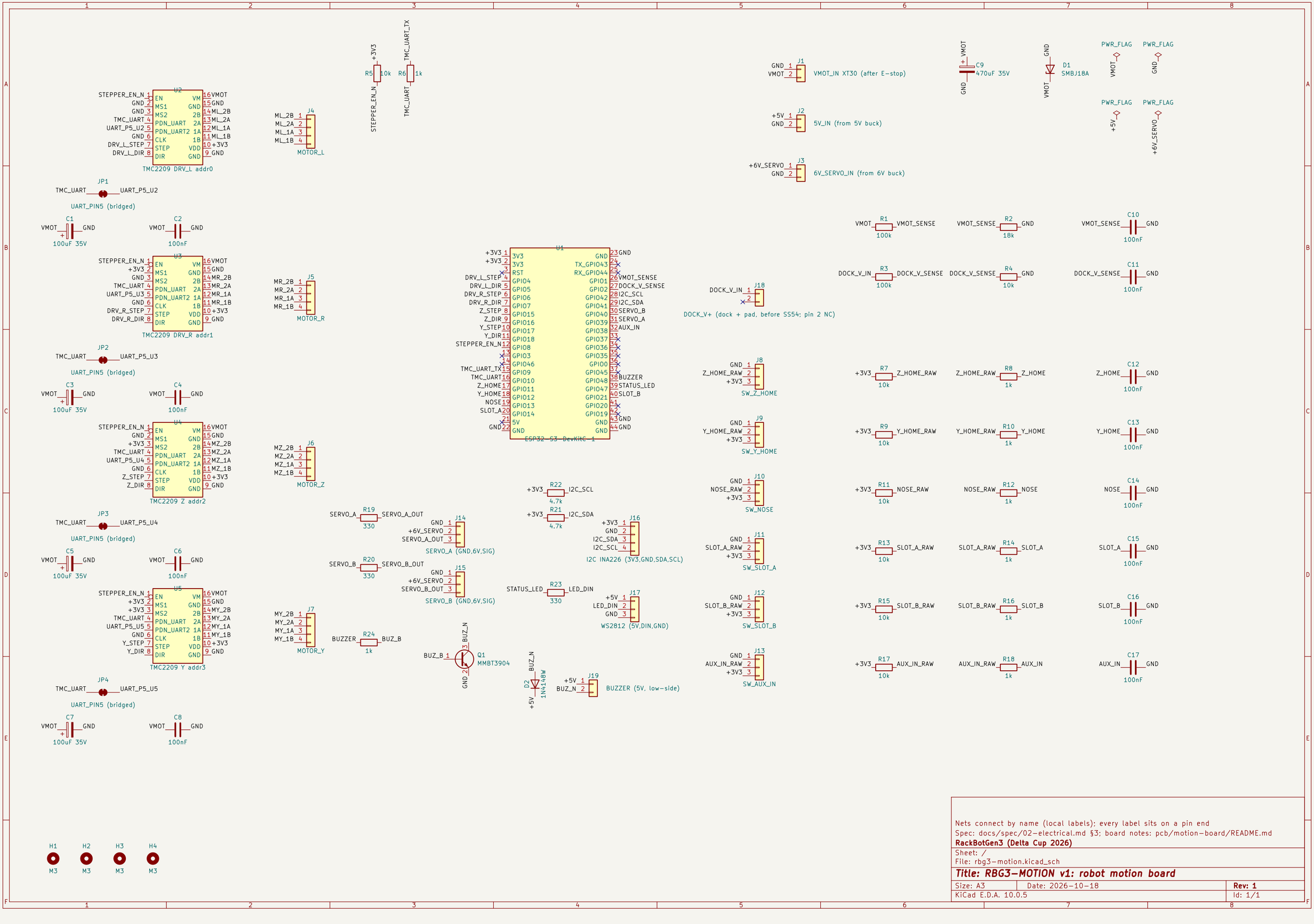
Out of scope (verified on the bought parts instead)

These are checked by measuring the parts when they arrive:

- the XL4015 module's shunt position;
- the XT30 polarity marks;
- the ESP32 header spacing;
- the relay coil current;
- the MKS TMC2209 R8 resistor.

They are listed in `docs/BACKLOG.md`.

Please reply with comments numbered to the questions, or as a GitHub PR review.



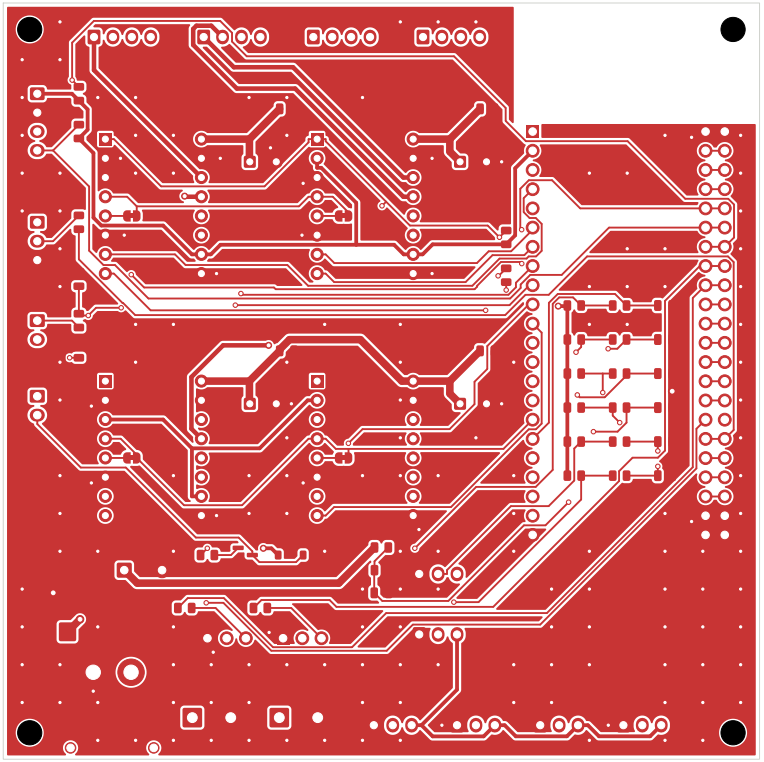
Nets connect by name (local labels); every label sits on a pin end
Spec: docs/spec/02-electrical.md §3; board notes: pcb/motion-board/README.md

RackBotGen3 (Delta Cup 2026)

Sheet: /
File: rbg3-motion.kicad_sch

Title: RBG3-MOTION v1: robot motion board

Size: A3	Date: 2026-10-18	**Rev: 1**
KiCad E.D.A. 10.0.5		Id: 1/1

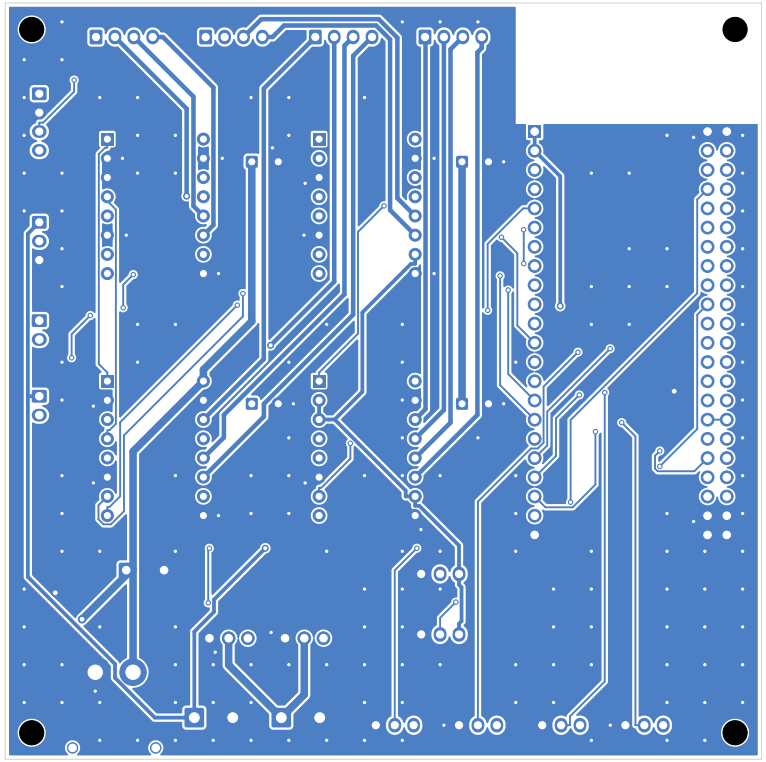


2 layers, 100 x 100 mm, 1.6 mm FR-4, 1 oz copper
Spec: docs/spec/02-electrical.md §3; board notes: pcb/motion-board/README.md
RackBotGen3 (Delta Cup 2026)

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KiCad E.D.A. 10.0.5		Id: 1/8

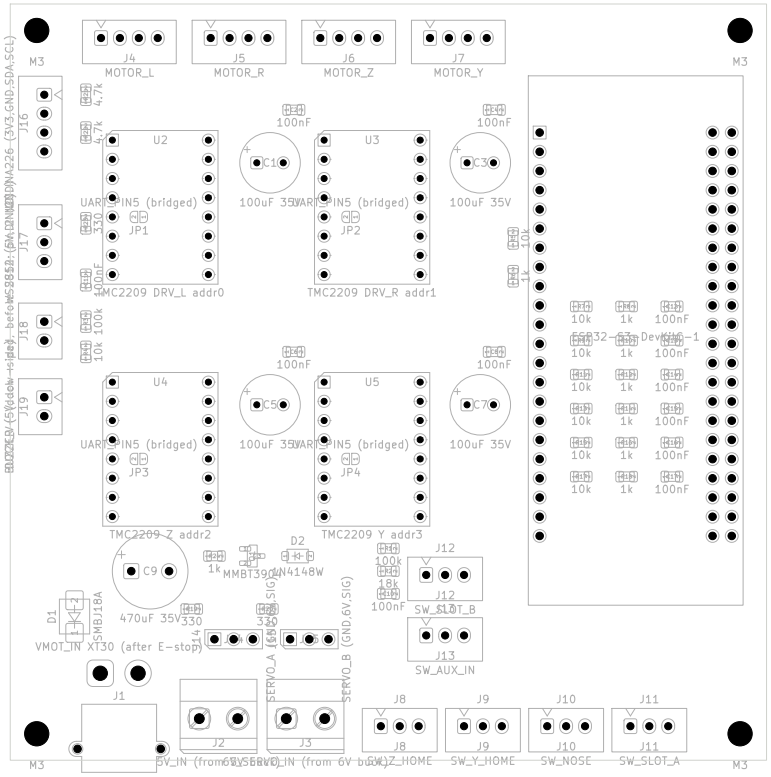


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Spec: docs/spec/02-electrical.md §3; board notes: pcb/motion-board/README.md
RackBotGen3 (Delta Cup 2026)

Sheet:
File: rbg3-motion.kicad_pcb

Title: RBG3-MOTION v1: robot motion board

Size: A4	Date: 2026-10-18	**Rev: 1**
KiCad E.D.A. 10.0.5		Id: 2/8

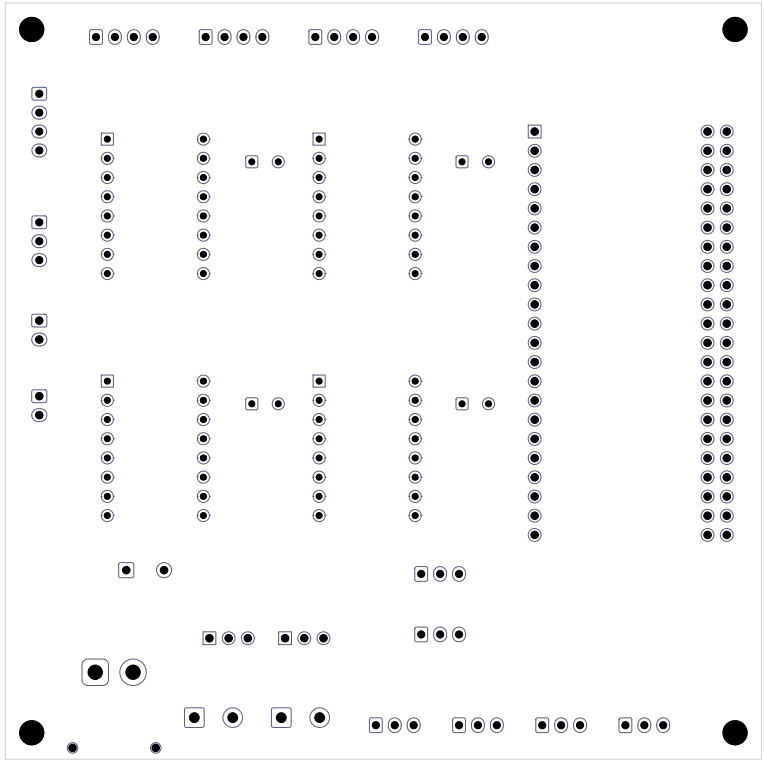


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Spec: docs/spec/02-electrical.md §3; board notes: pcb/motion-board/README.md
RackBotGen3 (Delta Cup 2026)

Sheet:
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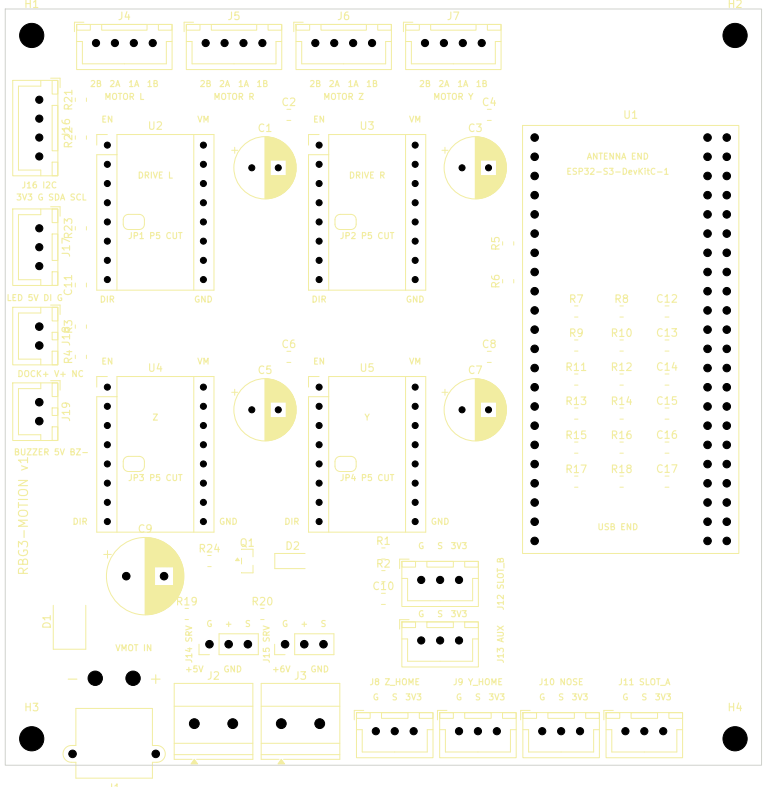


2 layers, 100 x 100 mm, 1.6 mm FR-4, 1 oz copper
Spec: docs/spec/02-electrical.md §3; board notes: pcb/motion-board/README.md
RackBotGen3 (Delta Cup 2026)

Sheet:
File: rbg3-motion.kicad_pcb

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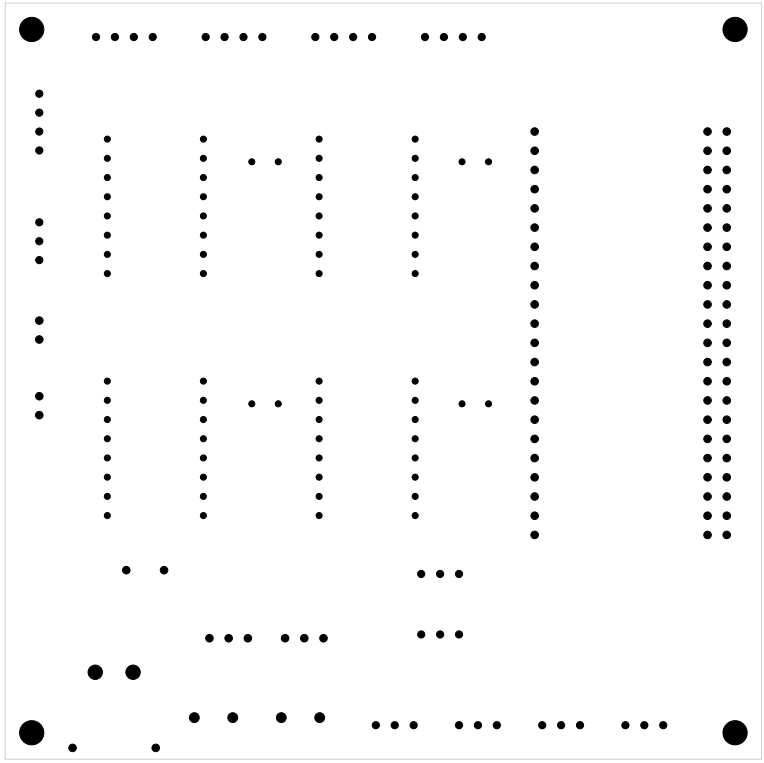


2 layers, 100 x 100 mm, 1.6 mm FR-4, 1 oz copper
Spec: docs/spec/02-electrical.md §3; board notes: pcb/motion-board/README.md
RackBotGen3 (Delta Cup 2026)

Sheet:
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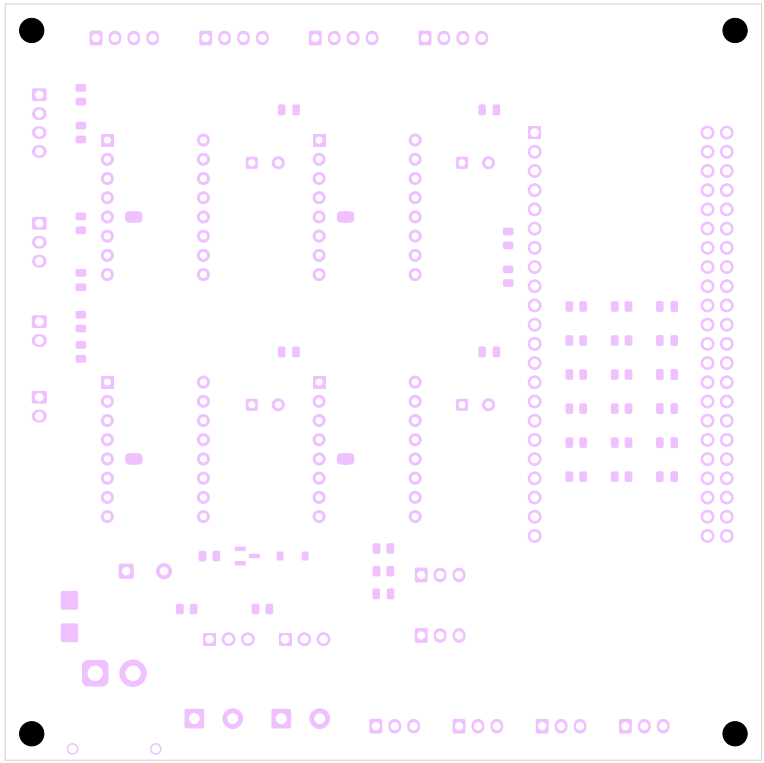


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RackBotGen3 (Delta Cup 2026)

Sheet:
File: rbg3-motion.kicad_pcb

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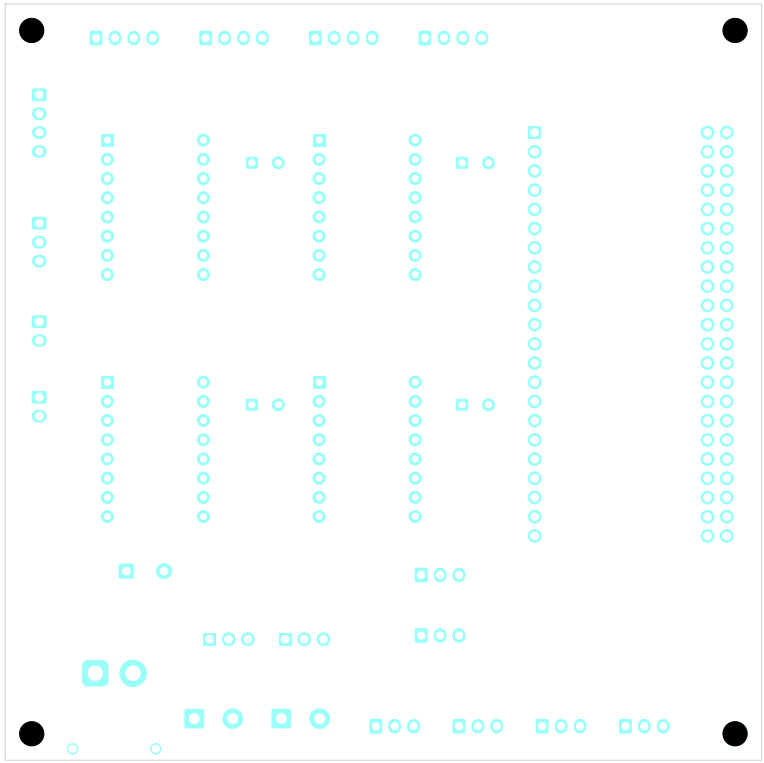


2 layers, 100 x 100 mm, 1.6 mm FR-4, 1 oz copper
Spec: docs/spec/02-electrical.md §3; board notes: pcb/motion-board/README.md
RackBotGen3 (Delta Cup 2026)

Sheet:
File: rbg3-motion.kicad_pcb

Title: RBG3-MOTION v1: robot motion board

Size: A4	Date: 2026-10-18	**Rev: 1**
KiCad E.D.A. 10.0.5		Id: 7/8



2 layers, 100 x 100 mm, 1.6 mm FR-4, 1 oz copper
Spec: docs/spec/02-electrical.md §3; board notes: pcb/motion-board/README.md
RackBotGen3 (Delta Cup 2026)

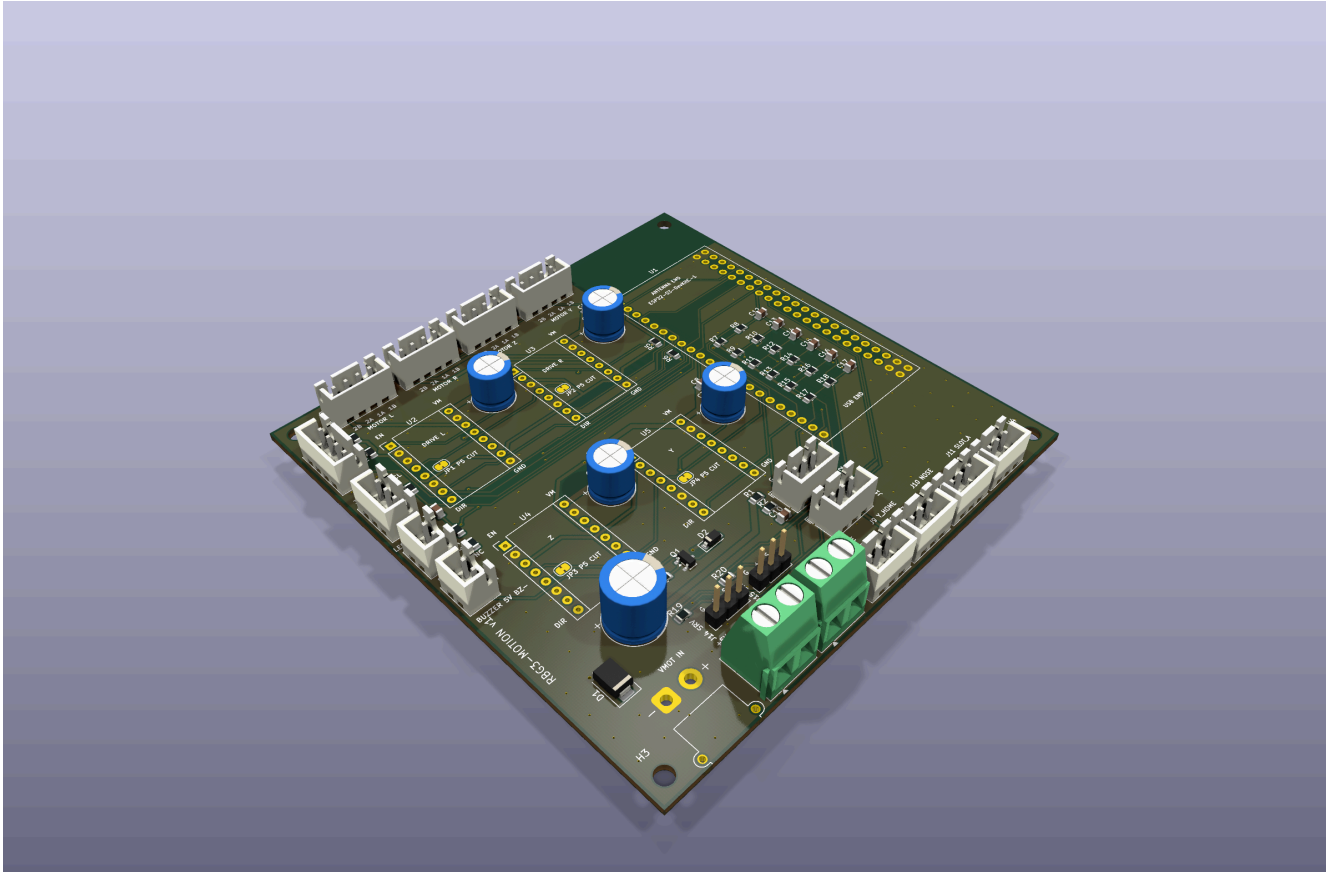
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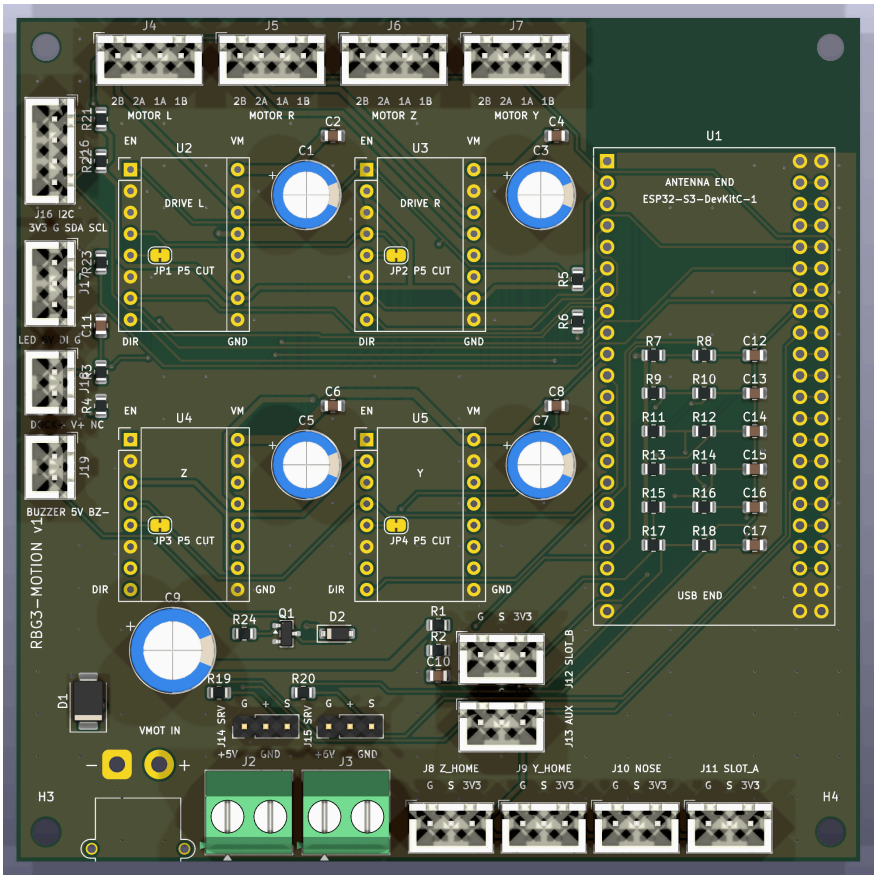
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3D renders

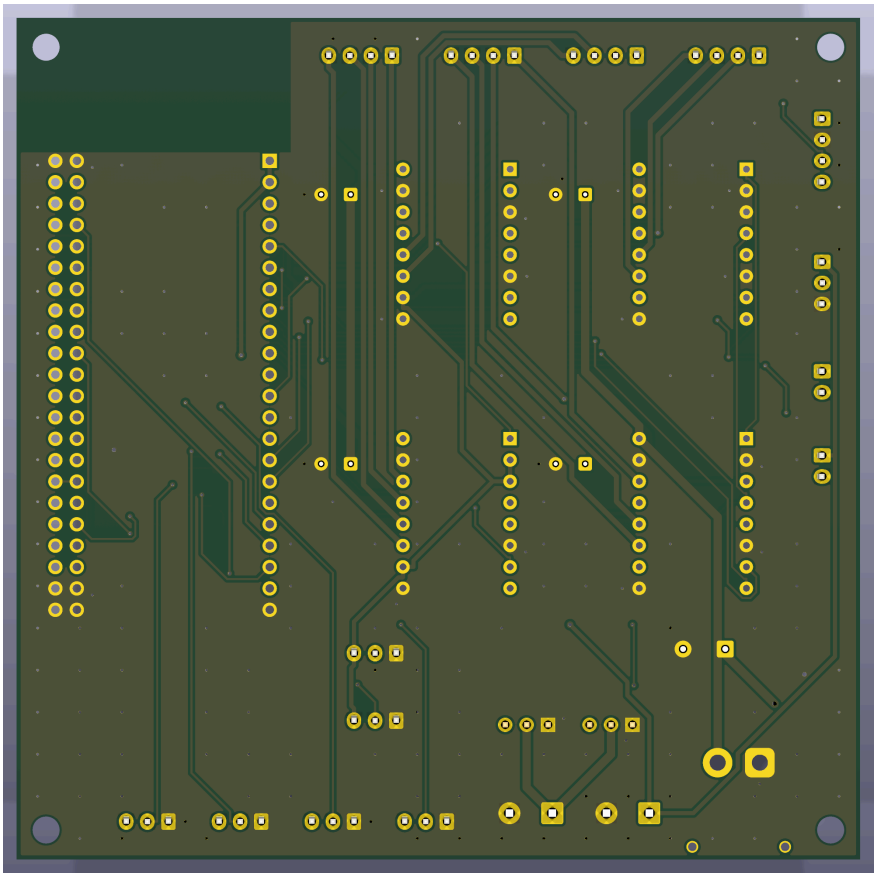
Rendered by KiCad from the board file. The ESP32 and TMC2209 modules plug into the empty sockets and are not drawn; J1 (XT30) has no 3D model.



pcb/motion-board/fab/rbg3-motion-3d-iso.png



pcb/motion-board/fab/rbg3-motion-3d-top.png



pcb/motion-board/fab/rbg3-motion-3d-bottom.png

Source: docs/spec/02-electrical.md

RackBotGen3 — Electrical Specification

Status: Draft v0.1. The motion board (§3) is the KiCad deliverable. Everything else uses off-the-shelf modules and wiring.

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1. Robot power

Item	Spec
Battery	**4S1P 18650**, 3.5 Ah cells (e.g. Samsung 35E / LG MJ1), 14.4 V nominal, 16.8 V full, 12.0 V cutoff → **50.4 Wh** (airline < 100 Wh)
BMS	4S, ≥ 15 A, common port (charge and discharge on the same P+/P–), balancing
Main fuse / switch	**7.5 A** blade fuse **first, at the battery P+** (so everything after it, INA226 included, is fused) + 10 A rocker switch at the battery box. 7.5 A sits under the Samsung 35E's 8 A max discharge (MJ1: 10 A); normal draw is < 2 A
E-stop	Robot mushroom (NC, twist release) in series with the coil of a **12 V automotive relay (SPST-NO, 30/40 A)** whose contact cuts **VMOT only** (steppers and servos). The button carries only the coil current (≈ 0.1–0.2 A), so its weak DC breaking rating does not matter ("10 A" on button listings is thermal current). Fail-safe: a pressed button, broken wire or loose connector drops the relay. 1N4007 flyback diode across the coil. The coil draws ≈ 1.5–3 W continuously (a few % of the battery per hour). The relay sits in the harness between the bus and J1 (no PCB change). Logic stays alive to report the E-stop. The Z lead screw is self-locking (dry/POM nut; marginal if greased), so the load cannot drop. Cutting VMOT resets the TMC2209 registers; firmware re-configures them before re-enabling (03 §8)
Energy meter	**INA226** module (**10 mΩ / R010** shunt, ±8 A; local modules ship with R100, so replace it with a 2512 R010 1 % shunt) between the fuse and the main bus. **+ current = discharge, – = charging**
Charging	Dock pads → Schottky (SS54) → **CC/CV buck (XL4015-type) set 16.8 V / 1.5 A** → main bus (feeds loads + charges through the BMS). Full charge ≈ 2.5 h; the dock also tops up between demo runs. **Charger grounding:** most XL4015 CC/CV boards sense current with a shunt between IN– and OUT–. Dock – connects **only** to the charger IN–, never to robot GND, or the current limit is bypassed (measure IN–↔OUT– on the bought module first). Lock the CV/CC trim pots with glue after setting and re-check before the event. The Pi turns dock power off (Modbus M0) once charge current tapers below 0.2 A
Dock sense	Dock **pad side, before the SS54** (so the charger cannot backfeed it) → divider → ESP32 ADC (`DOCK_V`, J18). Divider low side on robot GND (≈ 75 mV error from the charger shunt, negligible). **Docked = DOCK_V > 20 V** and PLC enabled dock power

```
flowchart LR
  BAT["4S Li-ion 50 Wh\n+ BMS"] --> F["Fuse 7.5 A\n+ main switch"] --> INA["INA226\nshunt"] --> BUS["VBAT bus\n12-16.8 V"]
  DOCK["Dock pads 24 V"] --> D["SS54"] --> CHG["CC/CV 16.8 V 1.5 A"] --> BUS
  DOCK -. "DOCK_V (pad side)" .-> ESP
  BUS --> ES["E-stop (NC)"] --> K["Relay coil 12 V\n(+ 1N4007)"]
  BUS --> KC["Relay contact (NO)\n30/40 A"] --> VMOT["VMOT"]
  K -. "closes" .-> KC
  VMOT --> TMC["4x TMC2209"]
  VMOT --> B6["Buck 5.8 V 3 A"] --> SERVO["2x MG90S"]
  BUS --> B5["Buck 5.15 V 5 A"] --> PI["Raspberry Pi 3\n(micro-USB pigtail)"]
  PI -- "USB (90° C plug)" --> ESP["ESP32-S3 DevKitC-1\n(powered over USB)"]
```

Power estimate: idle ≈ 4 W (Pi + logic), moving ≈ 12–20 W. **≈ 0.6–1.0 Wh per 3-min swap**, so a full battery lasts ≈ 2.5 h of continuous cycling.

Wire: battery/VMOT 18 AWG, motors 22–24 AWG, signals 26 AWG. Connectors: XT60 at the battery, XT30 to the board, JST-XH for motors and switches.

2. Compute & motion control

- **Raspberry Pi 3B/3B+**: Raspberry Pi OS Lite 64-bit, headless. IMX708 120° camera board (Camera Module 3 Wide class) on CSI (500 mm flex). USB to the ESP32 (serial). Wi-Fi client on the rack network.
- **ESP32-S3-DevKitC-1** (N8 or N16R8; **not** the -V 1.8 V variants, which put GPIO47/48 at 1.8 V) plugged into the motion board. **Powered over USB from the Pi** with a USB-A → USB-C cable whose C plug is **right-angled, cable exiting away from the board** (a straight plug reaches the switch connectors J10/J11 on the bottom edge). The DevKit 5V pin is *not* connected to the board 5 V (prevents backfeed).
- **4 × TMC2209 StepStick** (e.g. BTT v1.3), one shared single-wire UART, VIO from ESP32 3V3 (drivers disabled if the ESP32 is unpowered).

Driver	UART addr (MS1/MS2)	Motor	Run current (rms)	Microsteps
U2	0	Drive left	1.0 A	16 (interpolated)
U3	1	Drive right	1.0 A	16
U4	2	Z lift	0.9 A (hold 0.5 A)	16
U5	3	Y extractor	0.6 A	16

3. Motion board (KiCad PCB)

Two-layer module carrier, **100 × 100 mm**, 1 oz copper, 4 × M3 holes at 93 × 93 pitch. Ordered from JLCPCB **by 14 Oct**, with **Economic assembly of the 39 SMD parts on 2 of the 5 boards** (`fab/rbg3-motion-jlc-bom.csv` + `-jlc-cpl.csv`); the through-hole parts are hand-soldered. The same schematic can be built on perfboard as a fallback. **Implemented in** `pcb/motion-board/` (KiCad 10, ERC/DRC clean; see its README for the pre-order checklist). As built: VMOT/GND traces 1.0 mm + GND pours on both layers, motor traces 0.6 mm, +6 V servo 1.0 mm, other supplies 0.5 mm, signals 0.25 mm.

3.1 ESP32-S3 pin map

GPIO	Signal	Dir	Notes
4 / 5	DRV_L_STEP / DRV_L_DIR	out	
6 / 7	DRV_R_STEP / DRV_R_DIR	out	
15 / 16	Z_STEP / Z_DIR	out	
17 / 18	Y_STEP / Y_DIR	out	
8	STEPPER_EN_N	out	Shared, active low, 10 k pull-up to 3V3 (disabled at boot)
9	TMC_UART_TX	out	Through 1 k to the single-wire PDN_UART bus
10	TMC_UART_RX	in	Direct to the bus
1	VMOT_SENSE	ADC	100 k / 18 k divider (16.8 V → 2.56 V, readable to ≈ 19 V: the calibrated ADC range at 12 dB is 0–2.9 V). Below 9 V means E-stop pressed or VMOT off
2	DOCK_V_SENSE	ADC	100 k / 10 k divider (24 V → 2.18 V)
11	Z_HOME	in	**NC** contact to GND: closed (low) = not at home; open (high) = at home **or broken wire** (fail-safe)
12	Y_HOME	in	" (NC, fail-safe)
13	NOSE	in	Nose dock switch (both stations), NO to GND, active low
14	SLOT_A	in	Tray slot A presence

GPIO	Signal	Dir	Notes
21	SLOT_B	in	Tray slot B presence
39	SERVO_A	out	Finger A, 330 Ω series
40	SERVO_B	out	Finger B, 330 Ω series
41 / 42	I2C_SDA / I2C_SCL	i/o	INA226 @ 0x40, 4.7 k pull-ups
47	STATUS_LED	out	WS2812 (external), 330 Ω series
48	BUZZER	out	Via NPN (onboard RGB on some DevKit revisions also on 48/38: harmless)
38	AUX_IN	in	Spare switch input J13 (onboard RGB LED on some DevKit revisions shares 38: harmless as an input)
—	Avoid	—	0, 3, 45, 46 (strapping); 19/20 (USB); 43/44 (UART0); 33–37 (octal flash/PSRAM on R8/R16 modules)

Switch inputs: internal pull-up + 10 k external pull-up to 3V3, 1 k series, 100 nF to GND (debounce/ESD).

3.2 Connectors

Ref	Type	Pins
J1 VMOT	XT30PW-M (horizontal)	**pad 1 = GND (–), pad 2 = VMOT (+)** (after E-stop)
J2 5V_IN	2-pin 5.08 screw terminal	+5 V, GND from the 5 V buck (which also feeds the Pi). Not reverse-protected: polarity on the silkscreen
J3 6V_SERVO_IN	2-pin 5.08 screw terminal	+6 V rail (buck set to **5.8 V**), GND from the servo buck. Not reverse-protected: polarity on the silkscreen
J4…J7 (MOTOR_L/R/Z/Y)	JST-XH 4-pin	2B, 2A, 1A, 1B (match the StepStick pinout)
J8…J13	JST-XH 3-pin	GND, SIG, 3V3 (Z_HOME, Y_HOME, NOSE, SLOT_A, SLOT_B, AUX_IN on GPIO38). KW11 COM → GND, NC (home switches) or NO (others) → SIG; the 3V3 pin is spare
J14 / J15 (SERVO_A/B)	2.54 3-pin male header	GND, 6 V, SIG
J16 (I2C)	JST-XH 4-pin	3V3, GND, SDA, SCL
J17 (WS2812)	JST-XH 3-pin	5 V, DIN, GND
J18 (DOCK_V)	JST-XH 2-pin	1 = DOCK_V+ (from the dock + pad, before the SS54); **2 = not connected**. The divider is referenced to robot GND on the board, so only one wire is needed. There is deliberately no GND pin, because wiring dock – to robot GND would bypass the charger's current sense (§1)
J19 (BUZZER)	JST-XH 2-pin	5 V, low side (NPN)

3.3 On-board parts

- 2 × 22-pin female headers for the ESP32-S3-DevKitC-1. The right row is drilled at both 22.86 mm and 25.4 mm spacing; fit the header in the row that matches the measured module.
- 4 × StepStick sockets (2 × 8 female headers). MS1/MS2 **hard-wired** per socket for UART addresses 0–3 (no jumpers to forget). UART on StepStick pin 4 (PDN_UART); CLK tied to GND (internal clock). Each socket also has a **normally bridged solder jumper (JP1–JP4, copper strip)** from pin 5 to the UART bus, because **the MKS TMC2209 V2.0 (P17) has PDN/UART on pin 5** (pin 4 carries UART only if its optional R8 is fitted). With the jumper bridged, pin 4 and pin 5 both reach the bus, which also suits modules with UART on pin 4 and pin 5 not connected. Cut the strip only for a module that has another signal on pin 5. It sits between the socket rows, so it can only be reached with the module unplugged (labelled `JPn P5 CUT`).

- Per driver: 100 µF / 35 V electrolytic + 100 nF on VMOT. Bulk: 470 µF / 35 V at J1. **SMBJ18A TVS** right at J1 (unidirectional; standoff 18 V > 16.8 V, breakdown 20–22 V). It clamps switch and E-stop ringing at about 21–23 V, because those currents are small. The datasheet clamp of 29.2 V applies only at the full rated pulse current and is right at the TMC2209's 29 V operating limit (32 V absolute max). An SMBJ17A (27.6 V clamp) is the drop-in if more margin is wanted.
- Reverse polarity: the XT30 is keyed, so a reversed VMOT harness needs a mis-crimped pigtail. If that happens, the TVS conducts and the 7.5 A fuse blows, but until it does the fault current also returns through board GND, the 5 V buck wiring and the Pi↔ESP32 USB ground. Nothing on the board actively blocks reverse polarity (a series P-FET is a v2 option).
- Buck modules are **off-board** (vendor footprints vary): the 5 V/5 A buck feeds the Pi and J2 (5 V in), the servo buck (3 A, from VMOT, after the E-stop) feeds J3. It is **set to 5.8 V**: the MG90S is rated 4.8–6.0 V, so 5.8 V leaves margin for trim drift and ripple. The rail keeps its "6V" name on the board. The board has no on-board regulators.
- WS2812 level: driven from 3V3 through 330 Ω. Strictly out of spec (VIH = 0.7 × 5 V = 3.5 V) but it works with most parts on a short lead; it is a status light only. If it is unreliable, use a WS2812B-V5/SK6812 variant with a 3.3 V-compatible input. There is no buffer footprint on v1.
- Silkscreen (`pcb/motion-board/scripts/silk.py`): board name `RBG3-MOTION v1`, function and pin order at every connector (+5V/GND and +6V/GND at J2/J3, 2B 2A 1A 1B at J4–J7, G S 3V3 at J8–J13, G + S at J14/J15, and so on). Each StepStick socket has its module pin names at the corners (**EN / VM** at the top, **DIR / GND** at the bottom). A module plugged in rotated 180° puts VMOT on the EN line and destroys the driver and the ESP32. The DevKit socket is marked ANTENNA END / USB END.

3.4 KiCad deliverables

Schematic, PCB layout, DRC/ERC clean, Gerbers + drill + BOM + position file (JLCPCB format), 3D STEP export (imported into the robot CAD to check fit).

4. Rack electrical (Delta)

4.1 Parts

Item	Spec
Delta 24 V DIN-rail PSU	24 V, **≥ 75 W**, universal 100–240 VAC input (Thailand 220 V / 50 Hz). Chosen: **Delta LYTE II DRL-24V120W1EN** (24 V / 5 A, 120 W, 123.6 × 30 × 116.8 mm)
Delta DVP-12SE11T PLC	8 DI / 4 DO (NPN transistor), 24 VDC, built-in Ethernet, **Modbus TCP server**. Programmed with ISPSoft/WPLSoft
Dock power relay	24 V relay module (≥ 5 A contact) driven by PLC Y0, + 3 A polyfuse on the dock + line. Stays on the rack's DIN rail; its output runs to the charging dock at the **spare station** over a **2-core 0.75 mm² lead (≈ 1.5 m)**: ≈ 1.2 A at 24 V, < 0.1 V drop
Tower light	Mini 3-color 24 VDC LED (R/Y/G), common +
E-stop	Mushroom NC, 2 contacts: one to PLC X5, one in series with the dock relay coil (hardware interlock)
Router	Travel router (e.g. GL.iNet GL-MT300N-V2) powered from 24 → 5 V USB buck; LAN to PLC, Wi-Fi AP `RACKB0T`
Terminal blocks	DIN rail terminal blocks for 24 V / 0 V distribution

Power budget: PLC ≈ 5 W, light 3 W, router 3 W, robot charger ≈ 28 W → **≈ 40 W**, so the 120 W PSU runs at about a third of its rating.

4.2 PLC I/O map

PLC I/O	Signal	Notes
X0	Bay 1 caddy present	Lever switch, 24 V input
X1	Bay 2 caddy present	
X2	Bay 3 caddy present	
X3	Bay 4 caddy present	
X4	Spare-station dock switch	Robot nose fully docked at the charging dock (home). The rack dock has no switch; the robot's NOSE switch confirms docking there
X5	Rack E-stop OK	**NC**: 1 = OK, 0 = pressed
X6	Spare S1 (GOOD) present	4-core cable to the spare station (≈ 1.5 m): +24 V, X4, X6, X7
X7	Spare S2 (RMA) present	
Y0	Dock power enable	Relay → dock contacts at the spare station. **Interlock in ladder: Y0 = cmd AND X4 AND X5 AND watchdog OK**
Y1	Tower GREEN	Idle / all bays OK
Y2	Tower YELLOW	Swap in progress / degraded bay
Y3	Tower RED	Fault / E-stop / Pi watchdog lost

Input wiring: switches connect +24 V to Xn with S/S tied to 0 V (the DVP-SE inputs accept sink or source wiring through S/S; on above 15 V, 5 mA). Output wiring: lamp/relay coil between +24 V and Yn (NPN sinking). The 12SE11T transistor outputs need their own supply: **UP = +24 V, ZP = 0 V**. Ratings are 0.5 A per point and lamps ≤ 2.5 W per point. Flyback diode on the relay coil if the module has none.

The Modbus register map is in 03-software.md §4.

4.3 Network

Device	IP
Travel router (AP `RACKB0T`, WPA2)	192.168.8.1
Delta PLC	192.168.8.10 (static, set in ISPSoft/DCISoft)
Raspberry Pi	192.168.8.20 (static/DHCP reservation)
Dashboard clients (laptop/phone)	DHCP

4.4 Fallback rack controller (if no PLC)

ESP32 DevKit + 8-ch 24 V optocoupler input board + 4-ch MOSFET output board on the same DIN rail. The firmware exposes the **identical Modbus TCP register map** over Wi-Fi, so the Pi software does not change.

5. Safety summary

- Robot E-stop is a hard cut of VMOT. Rack E-stop is a hard cut of dock power, plus a PLC input → Pi stops the robot in software.
- PLC watchdog: if the Pi heartbeat register stops changing for 2 s → RED lamp and dock power off.
- Dock contacts (spare station) are energized only while the robot is fully docked there (X4) — no live 24 V exposed otherwise. The relay stays at the rack, so the rack E-stop still cuts dock power in hardware.

- Battery: BMS protection, fuse, XT60 removable. Never charge unattended. Charging is only via the CC/CV module.

Source: pcb/motion-board/README.md

RBG3-MOTION v1 — robot motion board

2-layer, 100 × 100 mm module carrier: ESP32-S3-DevKitC-1 + 4 × TMC2209 StepStick drivers, switch inputs, servo/LED/buzzer/I2C outputs, VMOT and dock voltage sensing. Spec: `docs/spec/02-electrical.md` §3.

File	What
`rbg3-motion.kicad_sch` / `.kicad_pcb` / `.kicad_pro`	KiCad 10 project (schematic ERC clean, DRC: 0 errors, 0 unconnected, 0 parity)
`lib/rbg3.kicad_sym`, `lib/rbg3.pretty/`	Custom symbols (`ESP32-S3-DevKitC-1`, `StepStick_TMC2209`) and the DevKit socket footprint
`scripts/build.sh`	Rebuilds the layout from scratch: text fix-ups → placement → silkscreen (`silk.py`) → GND pour → Freerouting → stitching vias → pours → DRC
`fab/rbg3-motion-gerbers.zip`	Gerbers + Excellon drill (separate PTH / NPTH files) for JLCPCB (2 layers, 1.6 mm, 1 oz, HASL is fine)
`fab/rbg3-motion-bom.csv`, `fab/rbg3-motion-pos.csv`	BOM and placement file (hand assembly expected: 0805 / SOT-23 / SOD-123 / SMB + through-hole)
`fab/rbg3-motion-jlc-bom.csv`, `fab/rbg3-motion-jlc-cpl.csv`	JLCPCB **Economic** assembly of the 39 SMD parts on 2 of the 5 boards (the plan since spec v0.1.20) (LCSC numbers in `scripts/jlc_pcba.py`; all basic or preferred, so no extended-part fees). Check Q1, D1 and D2 orientation in the placement preview
`fab/rbg3-motion-schematic.pdf`, `fab/rbg3-motion.step`	Schematic print and 3D board for the robot CAD

Rebuild: `scripts/build.sh` (needs KiCad 10 at `/Applications/KiCad`, Java 21+, Freerouting jar at `~/.kicad-mcp/freerouting.jar` or `$FREEROUTING_JAR`). Then `scripts/export_fab.sh` regenerates `fab/`.

Board summary

- **Power in**: J1 XT30 **VMOT** after the robot E-stop (**pad 2 = +, pad 1 = −**, matches the XT30 silkscreen), bottom-left corner, mating face 1.7 mm past the board edge so the plug goes in from outside, J2 **5 V** from the off-board 5 V/5 A buck, J3 **6 V servo** from the off-board 6 V/3 A buck. Bulk 470 µF + **SMBJ18A** unidirectional TVS right at J1 (cathode on VMOT: clamps ringing, and a reversed harness conducts and blows the 7.5 A fuse); 100 µF + 100 nF at every driver.
- **ESP32**: powered over USB from the Pi; its 5V pin is **not** connected. 3V3 from the DevKit feeds logic. Use a **90° USB-C plug** whose cable exits away from the board: the area under the USB end (y 179.5–191) is kept free of connectors (J12/J13 sit left of it), but J10/J11 on the bottom edge are ≈ 13 mm from the port.
- **Drivers**: UART addresses hard-wired with MS1/MS2: U2 drive-L = 0, U3 drive-R = 1, U4 Z = 2, U5 Y = 3. Single-wire UART on StepStick pin 4 (PDN_UART): ESP TX (GPIO9) via 1 k, RX (GPIO10) direct. CLK tied to GND. **JP1–JP4** (normally bridged solder jumpers between the socket rows, labelled `JPn P5 CUT`) join pin 5 to the bus, as the MKS TMC2209 V2.0 needs. Cut the copper strip only for a module with another signal on pin 5 (module unplugged).
- **Inputs** (J8–J13, JST-XH GND/SIG/3V3): Z_HOME, Y_HOME, NOSE, SLOT_A, SLOT_B, AUX — 10 k pull-up, 1 k series, 100 nF.
- **Outputs**: servos J14/J15 (GND, 6V, SIG, 330 Ω), WS2812 J17, buzzer J19 (low-side NPN), I2C J16 (INA226).
- **Sense**: VMOT 100 k/18 k → GPIO1 (16.8 V → 2.56 V), dock 100 k/10 k → GPIO2 (J18 pin 1; wire it to the dock + pad **before** the SS54, so the charger cannot backfeed it). J18 pin 2 is **not connected**: never wire

dock – to the robot, because it goes only to the charger IN–. Home switches (J8/J9) are wired NC for fail-safe homing.
- Net classes: VMOT, GND and +6V_SERVO 1.0 mm; motor 0.6 mm; +5V/+3V3 0.5 mm; signals 0.25 mm.
- GND stitching vias on a 5 mm grid, plus one next to every through-hole GND pad that has none nearby. Copper keep-out under the DevKit antenna (top-right corner).
- **Silkscreen** (`scripts/silk.py`): the board name, plus function and pin order at every connector. Each StepStick socket has its module pin names at the corners (EN/VM at the top, DIR/GND at the bottom), and the DevKit socket is marked ANTENNA END / USB END. The reference designators of J8–J15 and JP1–JP4 are on F.Fab; the function labels name those parts.

Check before ordering (things drawn from memory, not datasheets)

1. **DevKit header row spacing.** The right-hand row is drilled at **both 22.86 mm and 25.4 mm** from the left row. Measure your ESP32-S3-DevKitC-1 and solder the female header into the matching row only.
2. **DevKit pinout.** Compare `lib/rbg3.kicad_sym` (J1 = pins 1–22, J3 = pins 23–44, pin 1 at the antenna end) against Espressif's ESP32-S3-DevKitC-1 pin layout for your board revision.
3. **StepStick pinout.** Left column EN, MS1, MS2, PDN_UART, PDN_UART2, CLK, STEP, DIR; right column VM, GND, 2B, 2A, 1A, 1B, VDD, GND (Pololu layout). Check your TMC2209 module's silkscreen, especially pin 4 vs 5 for UART. JP1–JP4 come bridged (UART on pin 4 and 5); cut one only if that module uses pin 5 for something else. Insert each module so its EN pin matches the `EN` mark.
4. **Motor connector order** is 2B, 2A, 1A, 1B. If a motor runs backwards or stutters, swap one coil pair in the cable.
5. **Mounting holes**: M3 at 93 × 93 mm pitch (3.5 mm from each edge).

Assembly and bring-up checklist

Hand assembly (JLCPCB makes the bare board only). Tools: fine-tip iron, flux, 0.5–0.8 mm solder, tweezers, multimeter, current-limited bench supply.

Solder, low parts first so the board sits flat

JLCPCB-assembled boards: the first two items are already done; check them by eye and start at JP1–JP4. Only the 3 bare spares need them.

- ☐ 0805 resistors and capacitors (values in `fab/rbg3-motion-bom.csv`).
- ☐ Q1 MMBT3904 (SOT-23), D2 1N4148W (SOD-123, band as on the silkscreen), D1 SMBJ18A (SMB, **band toward VMOT**).
- ☐ JP1–JP4: check with a meter that each joins pin 5 to the UART bus (bridged on the board; nothing to solder for the MKS TMC2209 V2.0). Easiest before the sockets go on; always with the module unplugged.
- ☐ Through-hole, low to tall: J14/J15 servo headers → JST-XH (J4–J13, J16–J19) → StepStick sockets (8 rows) → DevKit female headers (**matching row only**, 22.86 or 25.4 mm) → J2/J3 terminals → J1 XT30 → electrolytics C1/C3/C5/C7/C9 last (**long leg to the + pad**).
- ☐ Clean the flux and look for bridges, especially at Q1, the socket rows and the DevKit rows.

Bring-up, stop at the first failure

- ☐ Nothing plugged in: no short from VMOT, +5V, +6V_SERVO or +3V3 to GND.
- ☐ Bench supply 16.8 V, limit 0.2 A, into J1 (pad 2 = +): idle current near zero, VMOT on all four VM pins, nothing warm. Don't test reverse polarity (by design it blows the 7.5 A fuse through D1).
- ☐ Set and check the 5.0 V and 5.8 V (servo, "6V" rail) bucks off-board before wiring them to J2/J3.

- [ ] ESP32 in, powered over USB (its 5V pin is not used): GPIO1 reads ≈ 2.56 V at 16.8 V; each input J8–J13 toggles (home switches are NC, so they read active until pressed).
- [ ] **VMOT off**, then fit the TMC2209s (EN pin at the `EN` mark) and connect the motors. Never plug or unplug a driver or a motor with VMOT on. Power up; the firmware sets motor current over UART. Check that addresses 0–3 answer, then jog each axis slowly (motor wrong way or stuttering: swap one coil pair in its cable).

Known cosmetic items

DRC reports 15 silkscreen-only items (reference designators touching each other or the C9 outline, and the J1 outline clipped by the board edge). JLCPCB clips silkscreen over pads automatically; tidy them in the KiCad PCB editor if wanted. It also reports 4 `lib_footprint_mismatch` warnings for JP1–JP4, whose courtyards are removed on purpose (`scripts/fix_text.py`).

Source: pcb/motion-board/sim/README.md

RBG3-MOTION: SPICE checks of the power path

These simulate the parts of the robot's power path that a layout check can't verify. The bought modules (XL4015 charger, BMS, TMC2209) have no SPICE models, so each one is reduced to a simple equivalent circuit. The assumptions are listed at the top of each `.cir` file.

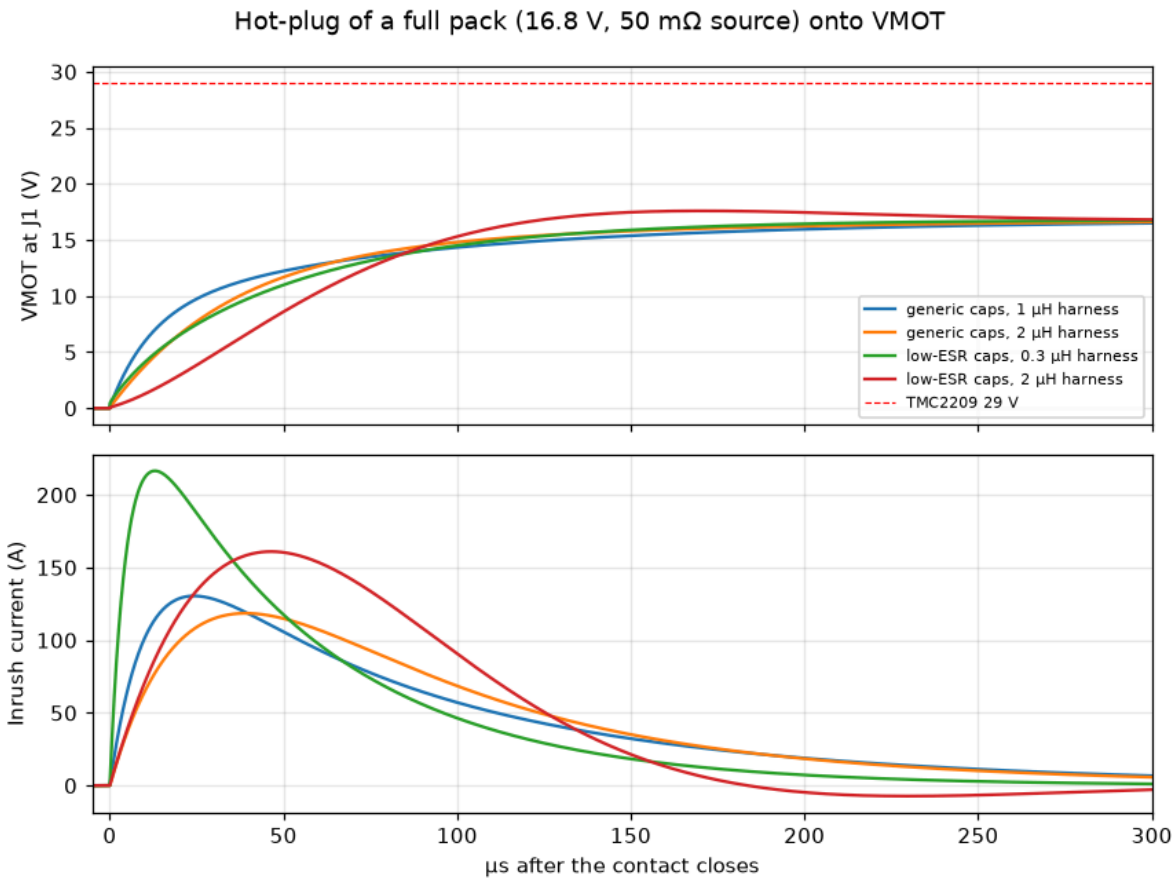
Run with `sim/run.sh` (from `pcb/motion-board/`). It uses the ngspice library bundled with KiCad 10, so nothing needs installing. It also works with an `ngspice` binary on PATH, or with `NGSPICE_LIB` set. It writes `results.md` (the full table) and `plots/`, and exits 1 if any check fails. Raw data goes to `sim/build/` (gitignored).

File	Circuit	Review question
`hotplug.cir`	A full pack switched onto the empty VMOT capacitors (XT30 plug-in, main switch, or E-stop relay release)	Q5, Q9
`estop.cir`	E-stop relay opens VMOT while all drivers carry current	Q4
`backfeed.cir`	Dock unpowered: the bus feeds back into the charger module and the dock pads	Q1
`signals.cir`	VMOT_SENSE / DOCK_V dividers vs the ADC range; single-wire UART through R6 1 k	Q10, Q11

Results (spec v0.1.19)

All checks pass. The details are in `results.md`.

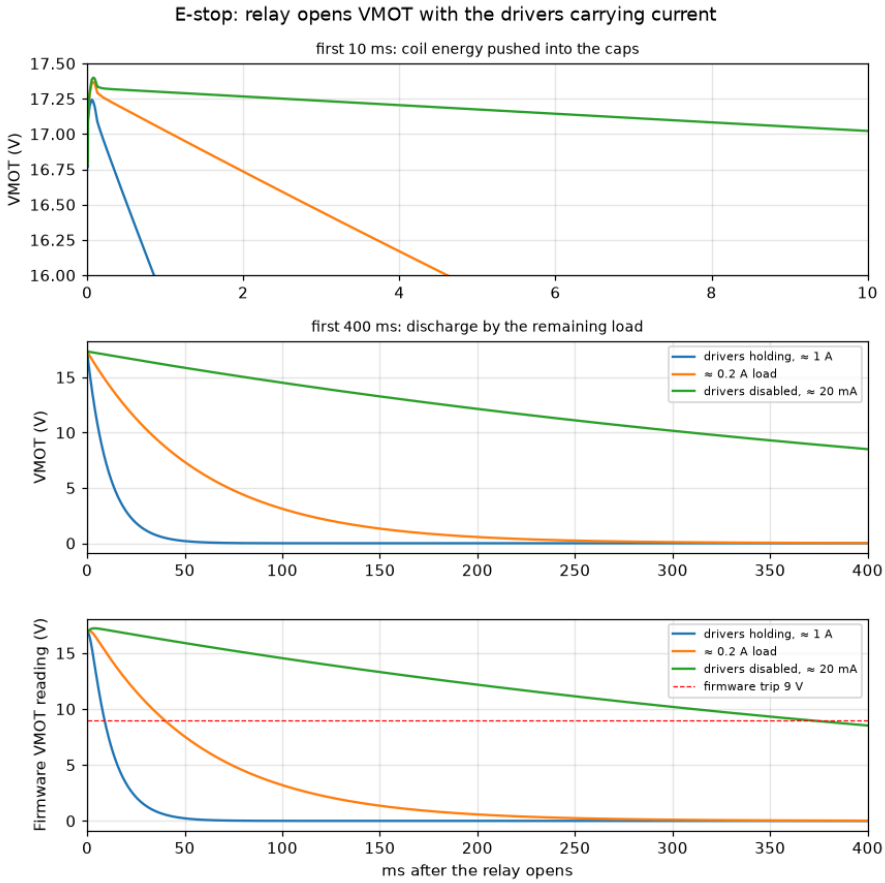
Check	Result	Limit
VMOT overshoot when a full pack is switched on	Overdamped: peak 16.8 V with generic capacitors, 17.6 V with low-ESR ones (2 µH harness). TVS never conducts	< 29 V
VMOT rise when the E-stop opens with all coils carrying current	+0.6 V (peak 17.4 V). TVS never conducts	< 29 V
Charger backfeed, DOCK_V false "docked"	DOCK_V reads at most 16.2 V	< 20 V
VMOT_SENSE / DOCK_V full scale	19.0 V / 31.9 V (2.56 V at 16.8 V, 2.18 V at 24 V)	≤ 2.9 V ADC
UART edge through 1 k	146 ns rise (1.7 % of a bit). A TMC reply reads 0.29 V low against the idle-high TX	ESP32 VIL 0.83 V

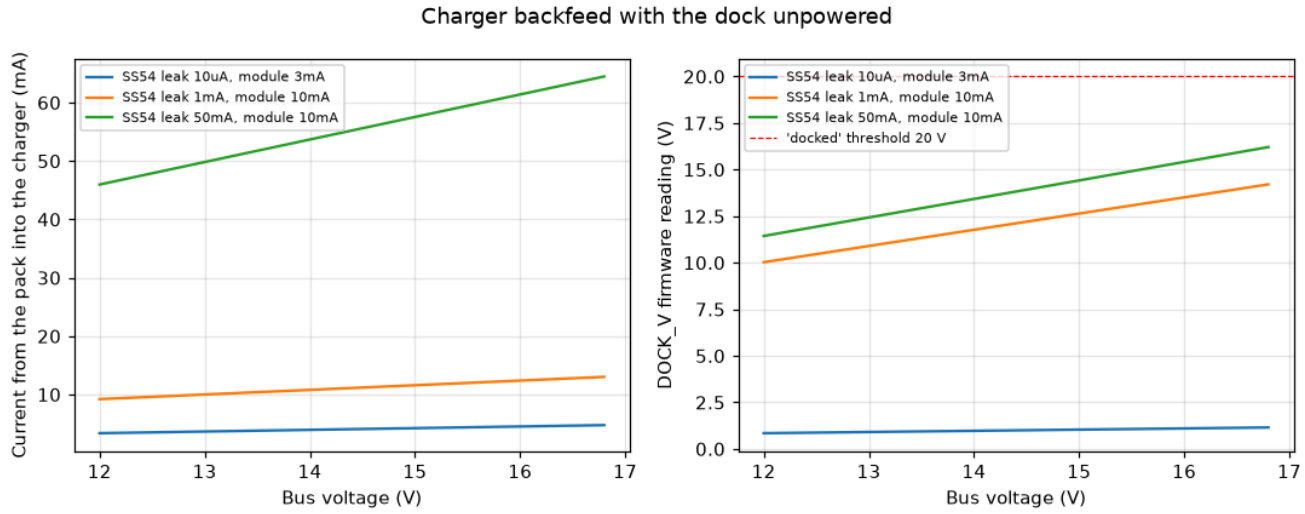


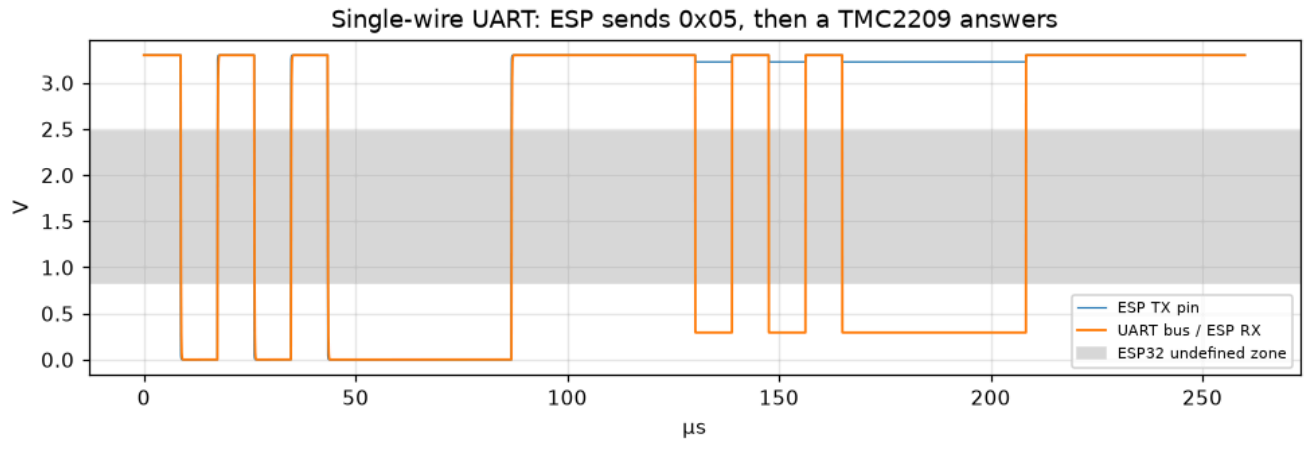
What the results mean

1. **Hot-plug: no overvoltage, but a big inrush.** The 870 µF of electrolytics and the 50–150 mΩ pack resistance make the circuit overdamped. The loop impedance √(L/C) is only 20–50 mΩ, so VMOT doesn't ring; it creeps up to 16.8 V. The SMBJ18A never conducts.
    - The inrush is 120–220 A for about 100 µs (I^2t ≈ 1–2 A^2s), every time the main switch closes or the E-stop is released.
    - Two things to **check on the bench**:
        - **The BMS short-circuit protection may trip.** Typical 4S BMS settings are 50–200 A with a 100–300 µs delay. If it trips at switch-on, add a pre-charge (an NTC or a resistor bypassed by the switch).
        - **The 7.5 A fuse** must have a melting I^2t comfortably above 2 A^2s. Check the datasheet of the fuse you bought.
    - This also argues **against swapping in low-ESR or ceramic bulk capacitors**: they raise the inrush and reduce the damping.
2. **E-stop: the coil energy is harmless.** The motors store about 7 mJ in total (L · I_{rms}^2, the four run currents). Dumped into 700 µF (870 µF at −20 %), that gives +0.6 V within 0.15 ms. The hand calculation gives 17.41 V; SPICE gives 17.40 V.
    - Adding the Z rotor's kinetic energy by hand (≈ 54 g·cm^2, typical for a 40 mm NEMA17, at 450 rpm, about 6 mJ) brings the worst case to about 17.9 V. That's still far below 29 V.
    - Back-EMF can't pump the bus. At 150 mm/s cruise the wheel motors turn at 36 rpm and make about 1 V. Even pushing the robot by hand at 1 m/s gives about 6 V, below the 12 V pack minimum. (Kt ≈ 0.41 N·m / 1.7 A ≈ 0.24 V·s/rad.)
    - **The E-stop is detected slowly when the drivers are disabled.** VMOT_SENSE < 9 V takes 9 ms with the drivers holding (≈ 1 A), 40 ms at 0.2 A, and about 370 ms with the drivers disabled (≈ 20 mA left on VMOT).

    - Safety isn't affected, because the motors lose power the moment the relay opens. But the backlog test "E-stop drops VMOT within 50 ms" only holds with the drivers enabled. Run test T10 that way, or add a load on VMOT if a faster report is wanted.
3. **Backfeed: no diode needed for correct function, but the pack is drained slightly.**
    - With the dock unpowered, the bus feeds the module's VIN side (worst case, through the body diode of the XL4015's internal switch) and its feedback divider. That draws **5–13 mA** (0.1–0.2 W, 2–5 % of the 4 W idle load).
    - Right after a charge the SS54 is hot. At its datasheet maximum leakage (50 mA at 100 °C) the drain rises to about 65 mA.
    - There's no drain with the main switch off, because the charger output is after the switch.
    - The dock pads can float up to about 16 V while undocked. That's harmless, since only the leakage current can flow.
    - **DOCK_V can't fake "docked":** the pad side can never rise above the bus (≤ 16.8 V), and the threshold is 20 V.
    - A series diode or ideal diode at the charger output would remove the drain. It isn't needed for correct function.
4. **The dividers and the UART are fine.** The values match CROSSCHECK, and the 1 k UART resistor adds a negligible RC delay.


E-stop: relay opens VMOT with the drivers carrying current
first 10 ms: coil energy pushed into the caps
VMOT (V)
first 400 ms: discharge by the remaining load
drivers holding, ≈ 1 A
≈ 0.2 A load
drivers disabled, ≈ 20 mA
Firmware VMOT reading (V)
firmware trip 9 V
ms after the relay opens


Charger backfeed with the dock unpowered
SS54 leak 10uA, module 3mA
SS54 leak 1mA, module 10mA
SS54 leak 50mA, module 10mA
Current from the pack into the charger (mA)
Bus voltage (V)
'docked' threshold 20 V
DOCK_V firmware reading (V)


Single-wire UART: ESP sends 0x05, then a TMC2209 answers
V
µs
ESP TX pin
UART bus / ESP RX
ESP32 undefined zone

What is not simulated

- The bought modules' behaviour: the XL4015's CC loop and stability, the BMS protection thresholds, and the TMC2209 chopper. All three are measured at bring-up (`../README.md`, Assembly and bring-up checklist).
- Relay contact bounce and arcing, and ESD on the switch and dock-pad lines.
- Trace heating (IPC-2152) and the PCB's own parasitics. Those are covered by the layout rules (1.0 mm VMOT, GND pours), not by these netlists.

Source: pcb/motion-board/sim/results.md

SPICE results (generated by `sim/run.py`, do not edit)

Case	Condition	Result	Limit	
Hot-plug	generic caps, 1 µH harness	VMOT peak 16.80 V (module pins incl.)	< 29 V	✓ pass
Hot-plug	generic caps, 1 µH harness	inrush 130 A peak, 108 µs above 50 A, I^2t 1.10 A^2s; TVS 0.0 mA	info	–
Hot-plug	generic caps, 2 µH harness	VMOT peak 16.80 V (module pins incl.)	< 29 V	✓ pass
Hot-plug	generic caps, 2 µH harness	inrush 119 A peak, 117 µs above 50 A, I^2t 1.10 A^2s; TVS 0.0 mA	info	–
Hot-plug	low-ESR caps, 0.3 µH harness	VMOT peak 16.80 V (module pins incl.)	< 29 V	✓ pass
Hot-plug	low-ESR caps, 0.3 µH harness	inrush 217 A peak, 95 µs above 50 A, I^2t 1.84 A^2s; TVS 0.0 mA	info	–
Hot-plug	low-ESR caps, 2 µH harness	VMOT peak 17.60 V (module pins incl.)	< 29 V	✓ pass
Hot-plug	low-ESR caps, 2 µH harness	inrush 161 A peak, 119 µs above 50 A, I^2t 1.83 A^2s; TVS 0.0 mA	info	–
E-stop	all coil energy into VMOT (fast decay), 16.8 V, C −20 %, any load	VMOT rise +0.60 V (peak 17.40 V), TVS 0.00 mA	< 29 V, TVS off	✓ pass
E-stop	drivers holding, ≈ 1 A: time to VMOT_SENSE < 9 V	9 ms	info (50 ms target, backlog)	–
E-stop	≈ 0.2 A load: time to VMOT_SENSE < 9 V	40 ms	info (50 ms target, backlog)	–
E-stop	drivers disabled, ≈ 20 mA: time to VMOT_SENSE < 9 V	370 ms	info (50 ms target, backlog)	–
Backfeed	SS54 leak 10uA, module 3mA, bus 16.8 V	pack drain 4.8 mA (0.08 W); pads 1.2 V; DOCK_V reads 1.2 V	DOCK_V < 20 V	✓ pass
Backfeed	SS54 leak 1mA, module 10mA, bus 16.8 V	pack drain 13.0 mA (0.22 W); pads 14.2 V; DOCK_V reads 14.2 V	DOCK_V < 20 V	✓ pass
Backfeed	SS54 leak 50mA, module 10mA, bus 16.8 V	pack drain 64.5 mA (1.08 W); pads 16.2 V; DOCK_V reads 16.2 V	DOCK_V < 20 V	✓ pass
ADC	VMOT_SENSE at 12.0 / 16.8 V; full scale	1.831 / 2.563 V; 2.9 V at 19.0 V	≤ 2.9 V at 16.8 V	✓ pass
ADC	DOCK_V at 24 V; full scale	2.182 V; 2.9 V at 31.9 V	≤ 2.9 V at 24 V	✓ pass
UART	ESP TX through 1 k into ≈ 65 pF bus	10–90 % rise 146 ns (1.7 % of a bit)	< 5 % of a bit	✓ pass
UART	TMC reply against idle-high TX (100 Ω driver assumed)	low 0.29 V, high 3.30 V	low < 0.83 V, high > 2.48 V (ESP32 VIL/VIH)	✓ pass

Source: docs/datasheets/CROSSCHECK.md

Datasheet cross-check (spec v0.1.15, 10 Oct 2026)

Spec, CAD and firmware values checked against the files in this folder. Page numbers are PDF pages. ✓ match · ✗ mismatch (fixed in the spec unless noted) · ⚠ open: no source, or a judgment call

Robot motion

Value (where)	Datasheet	Status
JK42HS40-1704: 1.7 A, 40 mm, ≥ 0.4 N·m (parts.csv P01)	1.7 A, 1.5 Ω, 2.3 mH, 4.2 kg·cm = 0.41 N·m, 0.28 kg (P01 catalog p. 32)	✓ (0.41 is only just ≥ 0.4)
5 mm D-shaft ≥ 22 mm, 31 mm hole pitch, Ø22 boss (`drive_bracket.py`, `z_motor_mount.py`, `rail_bracket.py`)	Ø5 × 24 ± 1, 31 ± 0.2, boss Ø22 × 2, M3 depth 4.5 (p. 32)	✓
Motor mass 280 g (`vendor.py` MASS_G)	0.28 kg (p. 32)	✓
"Torque ≈ 10 N per wheel" (01 §6.2)	0.41 N·m at the rated 1.7 A; the run current is 1.0 A → ≈ 0.25 N·m → ≈ 6 N at r = 40 mm (p. 32, p. 33 curve)	✗ → ≈ 6 N (still ≈ 4× the need)
Direct-drive wheel on the motor shaft (01 §6.2)	Max radial force **28 N at 20 mm from the flange** (p. 32). Static load ≈ 21 N per wheel with 2 loaded caddies (≈ 6 kg, 72 % on the axle), wheel centre ≈ 18 mm from the flange	⚠ 75 % of the limit before bumps. v0.1.19: hub bore no longer bottoms on the 24 ± 1 mm shaft (it was sized for 22 mm and would have pushed the wheel 1–3 mm out); hub fitted with a 2 mm shim
Z 15 mm/s on Tr8×2 = 450 rpm at 12 V (empty pack)	12 V / 1.6 A-peak curve ≈ 0.2 N·m at 450 rpm; Z needs ≈ 0.05 N·m (p. 33)	✓
TMC2209 VS limit "29 V" (02 §3.3, TVS note)	Operating 5.5–29 V; absolute max 32 V (P17_tmc2209 §20.1, §19)	✗ wording → 29 V operating / 32 V abs max
Run currents 1.0 / 1.0 / 0.9 / 0.6 A rms (02 §2, `settings.h`)	1.4 A rms design guideline per coil (§20.1)	✓
R_sense 0.11 Ω (`pins.h`)	MKS V2.0 schematic: R0805 0R11	✓
UART address = MS1 (bit 0), MS2 (bit 1)	§"UART": SLAVEADDR from MS1 bit 0, MS2 bit 1	✓
CLK tied to GND = internal clock	"Tie to GND using short wire for internal clock"	✓
Cutting VMOT resets the registers (02 §1, 03 §8)	§17: cycling VS resets the chip	✓
VIO from ESP32 3V3	VCC_IO 3.00–5.25 V	✓
UART on StepStick pin 4; JP1–JP4 open (02 §3.3)	**MKS TMC2209 V2.0 (P17): PDN/UART on pin 5**; pin 4 only if the optional R8 is fitted (P17 size-pin drawing)	✗ → JP1–JP4 are now bridged by default (v0.1.19)
MG90S supply (02 §3.3, P18)	Operating 4.8–6.0 V (P09 p. 1)	✓ buck set to 5.8 V (v0.1.18)
MG90S 22.8 × 12.2 × 28.5, 14 g (`robot_dims.py` SERVO)	22.5 × 12 × 35.5 (height probably includes the output spline), 13.4 g (P09 p. 1)	⚠ measure (already in the backlog)
MG90S stall current vs the 3 A servo buck	Not given	⚠ no source

Robot compute and I/O

Value (where)	Datasheet	Status
Strapping pins 0, 3, 45, 46 (02 §3.1)	Same (P16 datasheet, pin table notes)	✓
"35–37 (PSRAM)" avoided (02 §3.1)	Octal flash/PSRAM uses **GPIO33–37**	✗ → 33–37 (33/34 are unused anyway)
GPIO47/48 at 1.8 V on R8V/R16V	Same (pin table note 4)	✓
VMOT_SENSE on GPIO1, DOCK_V on GPIO2	ADC1_CH0 / ADC1_CH1 (not ADC2, so Wi-Fi safe)	✓
VMOT divider "readable to ≈ 20 V"	Calibrated ADC range at 12 dB is 0–2.9 V → 2.9 × 118/18 = **19.0 V** (Table 5-6)	✗ → ≈ 19 V (16.8 V reads 2.56 V ✓)
DOCK_V 24 V → 2.18 V	Within 0–2.9 V; readable to ≈ 32 V	✓
Onboard RGB on GPIO48/38	DevKitC-1 schematic: RGB_CTRL on GPIO38 (v1.1) / GPIO48 (v1.0)	✓
DevKit header row spacing 22.86 / 25.4	Not in the schematic	⚠ measure (already in parts.csv P16)
Pi 3B+ 85 × 56, holes 58 × 49 at 3.5 (`pi_tray.py`)	Same (P12 mechanical drawing)	✓
Pi 5 V input, buck 5.15 V / 5 A	5 V / 2.5 A (P12 brief)	✓
Camera ~120° diag, 102° H, focus ~150 mm, 25 × 24 (P13, 01 §6)	CM3 Wide: 120° diag, 102° H, focus 5 cm–∞, 25 × 24 × 12.4 mm (P13 brief pp. 4–5)	✓ for the genuine Wide; ⚠ the third-party IMX708 board must be measured
INA226 @ 0x40, ±8 A with R010, bus ≤ 16.8 V	A1 = A0 = GND → 0x40; ±81.92 mV / 10 mΩ = ±8.19 A; bus 0–36 V (P19 §5.5, Table 6-2)	✓
WS2812 VIH = 0.7 × VDD (02 §3.3)	0.7 VDD (PCB_ws2812b)	✓

Robot power

Value (where)	Datasheet	Status
35E 3.5 Ah, 4.2 V, 50.4 Wh (02 §1, P20)	3.6 V nominal, 3450 mAh typ / 3350 min, 4.20 V, cutoff 2.50 V → 4S = **49.7 Wh** (P20 Samsung p. 2)	✓ < 100 Wh; put the bought cell's Wh on the airline label
MJ1 3.5 Ah	3500 mAh, 3.635 V → 4S = **50.9 Wh** (P20 LG p. 3)	✓ same note
Charge 1.5 A (02 §1)	35E max charge 1.5 A (exactly at the limit); MJ1 standard 1.7 A, max 3.4 A	✓
10 A fuse, BMS ≥ 15 A	35E max discharge **8 A**; MJ1 10 A. Normal draw < 2 A	✗ → 7.5 A fuse (v0.1.19)
Pack cutoff 12.0 V	3.0 V/cell, above the 2.5 V minimum	✓ (conservative)
Servo buck MP2482 / SY8205 on VMOT ≤ 16.8 V, 3 A	MP2482 4.5–30 V, 5 A; SY8205 4.5–30 V, 5 A (P18)	✓
Charger XL4015 CC/CV 16.8 V / 1.5 A from 24 V	XL4015E1: Vin 8–36 V (abs max 40 V), Vout 1.25–32 V, 5 A, dropout 0.3 V, 7 A switch limit (P18_xl4015 pp. 1, 4–5)	✓ for 24 V → 16.8 V at 1.5 A. The chip has no CC loop: the 1.5 A limit comes from the module's own shunt + op-amp, so it can only be checked on the bought board (backlog: measure IN– ↔ OUT–)
XY-3606 5 V / 5 A (P18, P29 input ≥ 30 V)	No datasheet exists	⚠ check the listing / board on arrival
SS54 in the dock path	40 V, 5 A (PCB_ss54_mdd) vs 24 V / 1.5 A	✓

Value (where)	Datasheet	Status
SMBJ18A: standoff 18, breakdown 20–22, clamp 29.2 V; SMBJ17A 27.6 V	VWM 18, VBR 20.0–22.1, VC 29.2 V (Vishay, Littelfuse)	✓
VMOT relay coil 0.1–0.2 A, pulls in at 12 V, OK at 16.8 V (P38)	The Bosch file is a family overview with no coil data	⚠ no source (already in the backlog)
1N4007 flyback	1 A, 1000 V	✓

Motion board connectors and parts

Value (where)	Datasheet	Status
XT30PW-M on VMOT	15 A rated, 30 A for 1 min (PCB_xt30pw-m)	✓
XT30PW-M pad 1 = GND, pad 2 = VMOT	Pad numbering not shown in the sheet	⚠ check the footprint vs the real part's polarity marks
JST-XH for motors (1.0 A)	3 A with AWG 22 (PCB_jst-xh)	✓
MKDS 5.08 for 5 V / 5 A	10 A, 300 V (PCB_mkds)	✓
MMBT3904 buzzer driver	200 mA, 40 V	✓

Rack

Value (where)	Datasheet	Status
PSU 75–100 W (02 §4.1)	The saved example DRP024V060W1AZ is **60 W** (P24 p. 1)	✓ chosen DRL-24V120W1EN is 120 W (P24_delta-drl-24v120w1en); spec now ≥ 75 W (v0.1.19)
PLC ≈ 5 W in the power budget	DVP-12SE11T 1.5 W (P25 instruction sheet)	✓ (budget conservative)
Inputs: +24 V to Xn, S/S to 0 V ("verify sink/source")	S/S supports both SINK and SOURCE wiring; on at > 15 V, 5 mA	✓ (resolved)
Outputs Y0–Y3 NPN sinking	Transistor outputs need **UP = +24 V and ZP = 0 V** wired; 0.5 A/point, lamp ≤ 2.5 W/point	✗ UP/ZP missing from 02 §4.2 → added
Modbus M0 = 0x0800, D0 = 0x1000 (03 §4)	Not in the instruction sheet or catalogue (needs the DVP programming manual)	⚠ no source (verify during PLC bring-up)
12SE size	37.4 W × 90 H × 60 D, 35 mm DIN rail	✓

No datasheet at all (verify on purchased hardware; mostly already in `docs/BACKLOG.md`)

KW11-3Z, V-wheels, KFL08, Tr8×2 nut and coupler, GT2 pulley/idler, ball casters, 4S BMS, both E-stops, 24 V relay module, 3 A polyfuse, tower light current, dock contacts, travel router, third-party IMX708 board, CSI cable (only a 200 mm drawing exists). Estimates with no source: robot power 4 / 12–20 W, rack budget, MASS_G masses.

Open decisions (not changed in the spec)

1. ~~**Servo supply**~~: decided in v0.1.18, buck set to 5.8 V.

2. ~~**Drive shaft radial load**~~: decided in v0.1.19, wheel kept tight to the bracket plate (2 mm shim, bore clears the full shaft); outboard bearing only if the shafts show play.
3. ~~**Fuse vs cell rating**~~: decided in v0.1.19, 7.5 A main fuse (works with 35E and MJ1).